

Yifei Jiang

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Research Interests

Generative modeling, deep learning, neural computation, and machine learning for science, with applications to brain-computer interfaces, computational biology, and medical imaging.

Education

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| University of Michigan
Bachelor of Science in Engineering in Biomedical Engineering | 2023 – 2026
Ann Arbor, Michigan, United States |
| • GPA: 3.84/4.00. | |
| Shanghai Jiao Tong University
Bachelor of Engineering in Electrical and Computer Engineering | 2021 – 2026
Shanghai, China |
| • GPA: 3.65/4.00; rank: 56/230. | |

Research and Engineering Experience

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| SiClink
Algorithm Engineer | 2026 – Present
Shanghai, China |
| • Lead development of Image2MUA, the field's first image-conditioned generative model of multi-unit activity (MUA) in macaque visual cortex. | |
| • Integrated an attention-based mixture of visual encoders, sparse MoE routing, and rectified flow matching; achieved state-of-the-art accuracy of 0.87 on the macaque TVSD dataset; first-author manuscript submitted to ICLR 2027. | |
| University of Michigan
Research Assistant, Computational Immunology | 2023 – 2026
Ann Arbor, Michigan, United States |
| • Developed a cross-batch transcriptomic pipeline combining PLS-DA feature selection, support vector machines, and 1,000 bootstrap resamples for early identification of type 1 diabetes progressors. | |
| • Achieved 89% prediction accuracy under the study's bootstrap evaluation protocol; the predictions were used in downstream analyses of early pancreatic immune dysregulation. The work was released as a bioRxiv preprint and is under minor revision at the <i>Journal of Clinical Investigation</i> . | |
| Mayo Clinic
nuSURF Research Intern, Medical Computer Vision | Summer 2024
Rochester, Minnesota, United States |
| • Led architecture design, evaluation, and hyperparameter tuning for BWS-Net, a CNN-Transformer network with attention and adaptive skip connections for anterior bladder wall segmentation in low-contrast, noisy ultrasound. | |
| • Achieved a state-of-the-art Dice score of 0.82, advancing noninvasive bladder pressure assessment; published in <i>IEEE Journal of Biomedical and Health Informatics</i> (impact factor: 7.0). | |
| Beijing Genomics Institute (BGI)
Research Assistant, Computational Genomics | 2022 – 2023
Beijing, China |
| • Analyzed single-cell RNA-seq, bulk RNA-seq, and spatial transcriptomic data from medulloblastoma cohorts for biomarker and survival studies. | |
| • Conducted enrichment analysis, Cox regression, and cell annotation, contributing to published studies of tumor heterogeneity, cellular plasticity, and the immune microenvironment. | |

Selected Research Project

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| Property-Guided Catalyst Generation
Undergraduate Thesis | 2025 |
| • Built a property-conditioned diffusion model with a GNN denoiser, LoRA, and GRPO to generate task-specific catalysts from specified physical and chemical properties. | |
| • Reached a 19.53% condition-satisfaction rate, approaching fully fine-tuned SOTA performance; versus full-model training, improved efficiency 3×, cut compute cost by approximately 70%, and reduced GPU memory use to 30%. | |

Publications

Preprints and Manuscripts

- **Yifei Jiang**, M. Xu, S. Ma, and F. He. “Image2MUA: Image-Conditioned Generative Modeling of MUA with Visual-Expert Routing.” Manuscript under review, 2026.
- J. Roy, **Yifei Jiang**, et al. “Synthetic Immunological Niche Reveals Early Immune Dysregulation and Stratifies Therapeutic Response in Type 1 Diabetes.” *bioRxiv*, 2026; under minor revision at the *Journal of Clinical Investigation*. [DOI]
- D. Han, Z. Jia, Z. Wang, ..., **Yifei Jiang**, ..., and T. Jiang. “An Immune Microenvironment-Associated Gene Signature Predicts Outcomes and Therapeutic Response in Pediatric Medulloblastoma.” *Research Square*, 2024. [DOI]

Peer-reviewed Articles

- M. Saini, **Yifei Jiang**, et al. “BWS-Net: An Optimal Deep Learning Architecture for the Anterior Bladder Wall Segmentation Using Ultrasound Imaging.” *IEEE Journal of Biomedical and Health Informatics*, 2026. [DOI]
- J. Li, H. Liu, Z. Wang, ..., **Yifei Jiang**, ..., and T. Jiang. “Unveiling Spatial Heterogeneity in Medulloblastoma: A Multi-Omics Analysis of Cellular State and Geographical Organization.” *Neuro-Oncology*, 27(6):1611–1627, 2025. [DOI]
- H. Liu, J. Zhang, Z. Wang, ..., **Yifei Jiang**, ..., and T. Jiang. “High Cellular Plasticity State of Medulloblastoma Local Recurrence and Distant Dissemination.” *Cell Reports Medicine*, 6(1):101914, 2025. [DOI]
- J. Gao, Y. Zhao, Z. Wang, ..., **Yifei Jiang**, ..., and T. Jiang. “Single-Cell Transcriptomic Sequencing Identifies Subcutaneous Patient-Derived Xenograft Recapitulated Medulloblastoma.” *Animal Models and Experimental Medicine*, 8:458–472, 2025. [DOI]

Conference Presentations

- “Immune Biomarkers of Islet Transplant Rejection Revealed by Synthetic Immunological Niche.” Forthcoming oral presentation, BMES Annual Meeting, Orlando, Florida, 2026.
- “Synthetic Immunological Niche Reveals Early Immune Dysregulation and Stratifies Therapeutic Response in Type 1 Diabetes.” Forthcoming oral presentation, BMES Annual Meeting, Orlando, Florida, 2026.
- “Machine Learning-Based Early Prediction of Type 1 Diabetes from Immunological Niche Gene Signatures.” Oral presentation, TERMIS-Americas Annual Conference and Exhibition, Detroit, Michigan, 2025.
- “Deep Learning for Anterior Bladder Wall Segmentation in Ultrasound Images.” Poster presentation, KUH Summer Undergraduate Research Symposium, Charlottesville, Virginia, 2024.

Honors and Awards

- James B. Angell Scholar, University of Michigan, 2025.
- University Honors, University of Michigan, 2023 – 2025.
- Dean’s List, University of Michigan, 2023 – 2024.
- Honorable Mention, Interdisciplinary Contest in Modeling, 2022.

Teaching Experience

Shanghai Jiao Tong University

Teaching Assistant, General Chemistry

2022 – 2023

Shanghai, China

- Led discussion and problem-solving sessions for 30 students, designed and graded weekly assignments, and assisted with assessments for a course of approximately 200 students.

Technical Skills

- **Programming:** Python, R, C/C++ , MATLAB.
- **Machine Learning:** PyTorch, scikit-learn; generative models, diffusion and flow-matching methods
- **Additional Tools:** Mathematica, Verilog, Arduino.